

same and deflection will remain the same.

5(a)	Mass of copper in 1 m^3 of copper = 8960 kg Number of copper atoms in 1 m^3 of copper $= \frac{\text{total mass of all Cu atoms, } M}{\text{mass of 1 Cu atom, } m} \times N_A$ $= \frac{8960}{63.5 \times 10^{-3}} \times 6.02 \times 10^{23}$ $= 8.49 \times 10^{28}$ Since each copper atom has one conduction electron, the number of electrons as charge carriers in 1 m^3 of copper is $= 8.49 \times 10^{28}$
5(b)	$I = nAvq$ $\sqrt{\frac{P}{R}} = n \left(\frac{\pi d^2}{4} \right) v q$ $\sqrt{\frac{5.0}{30}} = (3.4 \times 10^{28}) \frac{\pi (0.36 \times 10^{-3})^2}{4} (1.6 \times 10^{-19}) v$ $v = 7.4 \times 10^{-4} \text{ m s}^{-1}$

5(c)	$R = \frac{\rho L}{A}$ When length is doubled but cross sectional area remains same, resistance doubles. $V = IR$ With same potential difference, current through wire is halved. $I = nA v_d q$ Hence drift velocity is halved.
6(a)(i)	$V = IR = (0.5 \times 10^{-3})(20) = 0.010 \text{ V}$